

CausalCache: Budget-Conditioned Behavioral Restoration for Long-Horizon GUI Memory

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Abstract

Long-horizon GUI agents must decide which historical visual evidence to expose to an action policy under a limited multimodal context budget. Existing memory controllers commonly learn from recency, similarity, attention, or salience labels, none of which directly measures whether high-fidelity evidence changes the policy’s executable decision. We introduce CAUSALCACHE, a policy-grounded supervision framework that upgrades one event at a time from a fixed-cost summary to its archived high-fidelity representation and measures how much the upgrade restores a validated frozen policy’s action behavior. We formulate a budget-conditioned, Shapley-style restoration value whose coalitions match the deployment budget, then distill the expensive offline values into a query-conditioned multi-budget gate. The method limits policy-visible context, not persistent storage, and filters reference states whose full-history action is not validated. We evaluate whether restoration supervision improves the closed-loop success—memory frontier and whether retained restoration value predicts success after controlling for next-action likelihood, context cost, and task horizon. **[Pending: Insert the principal AndroidWorld result and the matched-NLL effect size.]**

Introduction

A GUI trajectory can contain screenshots, actions, text arguments, coordinates, and state changes accumulated over dozens of decisions. Passing every event to a multimodal policy is expensive and can amplify context rot, while replacing all screenshots with summaries discards exact visual evidence. A memory controller must therefore decide which historical events deserve high-fidelity exposure at each decision.

Prior controllers select memory using recency, retrieval similarity, attention, learned salience, or task-state heuristics. These signals need not agree with the frozen action policy’s actual dependence on an event. Two memories can even have similar successor-action negative log-likelihood (NLL) while supporting different closed-loop outcomes. This motivates a narrower question: *how much does upgrading a historical event from a deterministic low-fidelity record to*

its high-fidelity visual evidence restore a validated policy’s executable action behavior?

CAUSALCACHE answers this question offline and uses the answers as supervision. It maintains a raw event archive, a cheap summary/index, and a limited policy-visible high-fidelity context. For validated decision states, it samples near-budget mixed-fidelity coalitions and estimates each event’s marginal behavioral restoration. The estimator is explicitly Shapley-style; our claimed contribution is the GUI intervention surface, the executable policy-behavior value, its alignment with the deployment budget, and closed-loop validation, not a new generic attribution estimator. A lightweight query-time gate then predicts these values from the current state, event summary, optional cheap visual embedding, age, and requested budget.

Our evaluation is organized around four falsifiable questions:

1. Does a restoration oracle improve executable task success at a fixed policy-visible visual budget?
2. Can a distilled gate approach the oracle without material online latency?
3. Does restoration supervision outperform the same gate architecture trained on next-action NLL or shuffled labels?
4. After matching or controlling for NLL, tokens, and horizon, does retained restoration value still explain terminal success?

Related Work

GUI memory. MementoGUI learns multimodal write, compression, and retrieval operators for a frozen GUI action model (Zeng et al. 2026). AndroTMem retrieves anchored intermediate states connected to the current subgoal (Shi et al. 2026). These systems establish the importance of learned controllers and dependency-critical events. CAUSALCACHE instead derives selector supervision from the marginal behavior change induced by restoring the high-fidelity version of the same event.

Interventional memory selection. CMI compares agent behavior under memory interventions (Srivastava 2026), while ATMem estimates how active memory states affect GUI task completion (Liu et al. 2026). Our scope is intentionally narrower: we hold the action policy fixed, intervene

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on low- versus high-fidelity GUI evidence, assign credit near the actual context budget, and validate the dense decision-time surrogate through executable closed-loop outcomes.

Shapley-style attribution. Permutation marginal contributions originate in cooperative game theory (Shapley 1953). We use this tool as an estimator, but change its value function and coalition distribution to reflect mixed-fidelity GUI context and a fixed operating budget.

Problem Formulation

Let the history before decision t be

$$H_t = \{e_1, \dots, e_{t-1}\}, \quad e_j = (o_{j-1}, a_j, o_j, \delta_j), \quad (1)$$

where δ_j is an optional deterministic UI delta. Every event has an archived high-fidelity representation e_j^{HF} and a fixed-cost low-fidelity record $\tilde{e}_j$. A query-time selector chooses the events exposed at high fidelity:

$$M_t \subseteq H_t, \quad \sum_{j \in M_t} c_j \leq B, \quad (2)$$

where c_j is the visual-token cost and B is the policy-visible context budget. Unselected events remain available as low-fidelity records. Raw events are retained in an external archive, so an event can be retrieved again at a later query; B is not a persistent storage or eviction budget.

The action policy π_0 is frozen. Only the selector is trained. Our goal is to improve executable task success across budgets without attributing gains to extra visual tokens or a stronger action policy.

CausalCache

Two-Level Event Representation

The deterministic low-fidelity record contains

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step_id, action_type,
target_text_or_coordinate_bin,
deterministic_ui_delta, result_status.
```

The archive stores the original screenshots, action, UI delta, and metadata. The gate may additionally use a precomputed cheap visual embedding z_j ; this avoids asking the summary alone to predict the importance of information that the summary necessarily omitted. The action policy receives mixed-fidelity inputs through a normal forward pass; we do not splice cached keys and values across incompatible contexts.

Validated Full-History Reference

For a full-history state, the frozen policy defines

$$q_t = \pi_0(\cdot \mid g, o_t, H_t). \quad (3)$$

We generate attribution labels only when the full-history canonical action a_t^* is validated by an executor-compatible reference action or belongs to a successful trajectory. We report reference coverage and compare against an unfiltered teacher. Thus, full history is a reference condition, not an assumed oracle.

Algorithm 1 Offline Budget-Conditioned Restoration Labels

Require: Validated state (g, o_t, H_t, a_t^*) , budget B , samples K

- 1: Run full history once to cache reference behavior q_t
- 2: **for** $k = 1$ to K **do**
- 3: Sample a shared near-budget coalition design for all events
- 4: **for** each event j evaluated by the design **do**
- 5: Run mixed-fidelity policy for S and $S \cup \{j\}$
- 6: Record $D_t(S) - D_t(S \cup \{j\})$
- 7: **end for**
- 8: **end for**
- 9: Return means, standard errors, and sample metadata

Executable Behavioral Distance

For a high-fidelity coalition S , all events outside S use their low-fidelity records. We measure pathwise divergence along the canonical action rather than claiming a full sequence-action KL:

$$D_t(S) = \alpha D_{\text{type}}(S) + \beta D_{\text{target}}(S) + \gamma D_{\text{text}}(S). \quad (4)$$

Action type uses teacher-forced token divergence; coordinates are mapped to UI elements or coarse bins; equivalent swipe grammars are mapped to viewport scroll direction; text arguments are normalized before comparison. Executable action match is the primary action metric, with raw token KL as an auxiliary analysis.

Budget-Conditioned Restoration

For event j , let $\mathcal{C}_B(j)$ be a distribution over feasible coalitions $S \subseteq H_t \setminus \{j\}$ that leave room for c_j and are near budget. The budget-conditioned restoration value is

$$G_{j,t}^{(B)} = \mathbb{E}_{S \sim \mathcal{C}_B(j)} [D_t(S) - D_t(S \cup \{j\})]. \quad (5)$$

When all high-fidelity blocks have equal cost, $\mathcal{C}_B(j)$ is uniform over coalitions of size $B - 1$. Unlike a standard Shapley value averaged over all coalition sizes, Equation 5 asks whether restoring j helps when the system already operates near its deployed budget. We use shared, antithetic, or stratified coalition samples where applicable and report standard errors, ranking stability, and top- B overlap across seeds.

Multi-Budget Query-Time Gate

The distilled selector predicts

$$s_{j,t} = f_\theta(g, o_t, \tilde{e}_j, z_j, \Delta t, B) \quad (6)$$

with regression, pairwise-ranking, and budget-aware set losses. At inference, the selector solves a cost-constrained positive-value selection problem:

$$M_t = \arg \max_{M: \sum_{j \in M} c_j \leq B} \sum_{j \in M} \max(s_{j,t} - \tau, 0). \quad (7)$$

For equal-cost event blocks this reduces to selecting up to, rather than exactly, the top B events above threshold τ . This permits a null decision when all candidate restorations are predicted to be harmful or useless.

Experimental Protocol

Benchmarks and Policies

AndroidWorld (Rawles et al. 2025) is the sole primary closed-loop benchmark. GUIOdyssey (Lu et al. 2025) or an equivalent long-horizon mobile dataset provides offline attribution and action analysis. We use one primary frozen policy and one transfer backbone. Every comparison shares the same low-fidelity channel, action policy, gate architecture where applicable, and visual-token accounting.

Comparisons

The compact main comparison includes summary only, recent, similarity, the same gate trained on next-action NLL or salience, the strongest reproducible GUI memory controller, a restoration oracle, and distilled CAUSALCACHE. A random label shuffle uses the same training states and gate capacity as a negative control.

Success–Memory Frontier

We report closed-loop task success against visual tokens and end-to-end latency, plus success per unit memory/compute. Restoration must improve the Pareto frontier rather than consume additional context. **[Pending: Insert main frontier plot and paired confidence intervals.]**

Matched-NLL Mechanism Test

Within task, budget, and horizon strata, we match memory decisions by successor NLL and test whether retained restoration mass remains associated with terminal success. The preregistered conditional model is

$$\Pr(y = 1) = \sigma(\beta_0 + \beta_1 R + \beta_2 \text{NLL} + \beta_3 \text{tokens} + \beta_4 \text{horizon}), \quad (8)$$

where R is retained restoration mass. The mechanism claim requires a stable $\beta_1 > 0$ under paired bootstrap confidence intervals and reasonable matching tolerances. We will report all eligible pairs rather than selected anecdotes.

Attribution Cost and Stability

We vary coalition samples K and report attribution standard error, Spearman rank correlation to a high- K estimate, top-budget Jaccard overlap, oracle success, total offline labeling compute, and online selector latency. The per-permutation telescoping identity is used only as an implementation sanity check. Coalition reconstruction error measures non-additive interactions.

Limitations and Claim Boundary

Restoration value is a dense decision-time surrogate. It is not a direct long-horizon causal effect, and its relevance to long-horizon control must be established through closed-loop success and the matched-NLL analysis. The causal language in this paper refers only to a controlled low-/high-fidelity intervention on a frozen policy’s behavior. We do not identify environment-level structural causality and do not learn a world model. The method should be weakened or abandoned if restoration fails to beat recency, similarity, and attention;

if its matched-NLL association disappears; if the distilled gate cannot approach the oracle; or if gains come from extra tokens.

Conclusion

CAUSALCACHeturns mixed-fidelity behavioral restoration into policy-grounded supervision for GUI memory selection. Its central empirical burden is not that attribution can fit a selector, but that budget-aligned restoration identifies high-fidelity evidence that improves executable long-horizon control beyond what next-action likelihood and context size explain.

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